

The Minoan receipt is native: the form of bookkeeping in six Bronze Age administrations

Alberto Acedo Biome Makers Inc. Draft v0.2, 9 September 2026. The transaction terms are credited to Schoep 2002; the Ugarit and Alalakh limitation is declared. Not for circulation.

Abstract

Bronze Age administrative archives share an anatomy: a heading, entries pairing a party or a commodity with a quantity, and totals. Within that anatomy each tradition chooses which functions to put into words. Six archives are measured here with the same procedure: Linear A at Hagia Triada (Crete, c. 1450 BC), the Middle Assyrian archival texts (14th-11th c. BC), the Amarna letters (14th c.), Ur III Sumerian (c. 2100-2000), Proto-Elamite Susa (c. 3000) and proto-cuneiform Uruk (c. 3350-3000), with Neo-Assyrian SAA 7 (7th c.) as a late control. For each, the closed class of recurrent functional words is extracted and assigned to five functions: totals, transfer, responsibility and sealing, date, and measure. All six verbalise totals and measures. The Mesopotamian and Assyrian archives verbalise transfer and date with the highest-frequency words of their corpora: in Middle Assyrian, “to give” and “to take” account for over a thousand instances and the year eponym appears 1,849 times, in a corpus of 89,677 lemmatised instances. Linear A verbalises neither: no word in its vocabulary behaves as a verb of transfer, and none behaves as a date, while the same corpus has four terms for totals and deficits fixed by arithmetic. The absence is systematic, not a gap in identification. Since the shared anatomy also holds for Proto-Elamite, which had no contact with Crete, and since the two functions Crete lacks are precisely the ones that define the Near Eastern model, the conclusion is that Minoan bookkeeping is not an adopted form: Crete imported commodities and their words, not the way of keeping accounts. Its receipt rests on the sealed object and on an implicit annual cycle instead of on the verb and the written date, which is consistent with what archaeology reports of Minoan archives.

The identification of those terms is not a contribution of this work. Schoep (2002) established them with their functions: TE, separated by dividers on both sides and always at the head of a list with one or more logograms; KU-RO, “total”, followed as in Linear B by the sum of the preceding entries; PO-TO-KU-RO, “grand total”; KI-RO, only at Hagia Triada, as a deficit, on the argument that when it follows a KU-RO it is always smaller than it; and SA-RA₂, also exclusive to Hagia Triada, whose mutual exclusion with TE within the mixed-commodity tablets suggests that the two represent opposed transactions, contributions against distributions or payments against rations. What is contributed here is not the inventory of terms but the comparison of functional repertoires across six archives: which administrative functions each tradition puts into words, and which it does not.

Keywords: Linear A, Middle Assyrian, Ur III, Proto-Elamite, administrative typology, sealing, Bronze Age trade.

1. The question

That Minoan writing was born for bookkeeping is not disputed; whether its form is native or borrowed is. Crete traded with the Levant and with Egypt from the Protopalatial period, its products appear at Ugarit and at Amarna, and commodities reached Crete from the East whose names Greek later inherited through Minoan (sesame, cumin). The transmission of goods and of words is documented. Was the form of the document transmitted too?

The question is answerable without reading Minoan, because the form of an archive can be measured: which administrative functions it puts into words, and how often. This paper measures that in six archives, and the answer is no.

2. Method and data

For each corpus the closed class is extracted: the lexical units that recur across many documents and that are neither proper names nor commodities. In the lemmatised corpora (Middle Assyrian, Amarna, Neo-Assyrian) the class is read directly from the glossary, with its instance counts; in the unlemmatised ones (Ur III, Proto-Elamite, Uruk, Linear A) the forms taken are those occurring in a minimum number of documents and, where more than one site is available, at several sites. Each term is then assigned to one of five administrative functions: total and account; transfer (to give, to receive, to issue, by the hand of); responsibility and sealing; date; measure. The assignment is the field's own in the read corpora, and ours by distribution in Linear A, where three of the terms are fixed additionally by arithmetic.

Corpora: Linear A (GORILA via the LinearA Explorer, 1,098 word occurrences on tablets, 782 types); TCMA, Archival Texts of the Middle Assyrian Period (ORACC, 2,276 tablets, 89,677 lemmatised instances, 2,524 lemmas); The Amarna Letters (ORACC, 305 texts, 29,961 instances); SAA 7 (ORACC, 219 texts, 8,574 instances); BDTNS for Ur III (20,000 tablets sampled from 101,389, 432,915 tokens); CDLI for Proto-Elamite (1,594 tablets) and Uruk IV-III (6,387).

3. The anatomy is common

The six archives write the entry as unit plus quantity, open with a heading and close with a total on part of the documents. The proportion of entries with a quantity ranges from 0.60 to 0.80. The partition of the lexicon is the same in all: a short class of frequent, repeated terms, which are the commodities and the functions, and a long class of unique units, which are the names and the places. At Susa, one-sign strings are singletons in 3% of cases and strings of three or more in 87-99%; at Haghia Triada 76% of the vocabulary is singleton; in Ur III and in Middle Assyrian the proportion is of the same order once proper names are discounted, which both corpora mark explicitly.

That anatomy holds for Proto-Elamite and for Uruk, separated from Crete by fifteen hundred years and two thousand kilometres, with no relation between Susa and Crete. It is the shape a record of deliveries takes when it is written, and it requires no transmission.

4. The five functions, and the two absences

function	Middle Assyrian (14th-11th c.)	Amarna (14th c.)	Ur III (c. 2000)	Linear A (c. 1450)
total and account	napharu “total” (811), nikkassu “account” (93)	gabbu “totality” (184)	šu-nigin ₂ , nig ₂ -ka ₉	KU-RO (37), PO-TO-KU-RO, DA-I
deficit	(marked by subtraction)	—	la ₂	KI-RO (16)
transfer	nadānu “to give” (613), nadnu “given” (401), leqû “to take” (365)	nadānu (238), leqû (213)	ba-ti “received”, ba-zi “issued”, giri ₃ “by the hand of”	none
responsibility and sealing	kunukku “seal” (789), ʔupšarru “scribe” (391), waklu “overseer” (88)	ʔupšarru (4)	kišib, maškim, ugula	the physical seal; TE on the nodules; A-DU and KA-PA as headings
date	ūmu “day” (2,599), līmu “year eponym” (1,849), warhu “month” (251)	ūmu (50), šattu (33)	mu, iti, u ₄	none
measure	sūtu (4,459), imēru (4,046), qû (1,779)	mina, shekel (residual)	gur, sila ₃ , gin ₂	its own system of fractions

Two observations on the table. The first concerns coverage: the twenty most frequent units cover 47.6% of the Middle Assyrian corpus, 52.2% of Amarna, 28.3% of Ur III and 13.7% of Linear A. Crete’s low figure does not indicate a poorer functional vocabulary but that there commodities are written with logograms and not with words: the space that lexicon occupies in Assyria is occupied in Crete by commodity signs and quantities. Comparing like with like, the size of the closed class is of the same order in all six: between fifteen and twenty-five terms.

The second is the one that matters. In the nearest contemporary archive, transfer and date are not marginal functions: “to give” and “to take” amount to over a thousand instances, the year eponym appears 1,849 times and the day 2,599, that is, nearly every tablet is dated. In Linear A there is no word that behaves as a verb of transfer or as a date. And the absence is not one of identification: of the twenty words that make up the closed class of Linear A, all have a function

assigned by position (totals, headings, anchors of the libation formula) or are frequent names; no candidate is left without a destination that could be a “received” or a “year”.

5. What Crete does instead

It delegates the two functions to the object and to the cycle.

The seal does the work of the verb. Haghia Triada produced hundreds of nodules and roundels against a modest number of tablets; the roundel carries a sign-group, sometimes a commodity logogram, and as many seal impressions as units. It is a complete receipt without any word of transfer: who delivers is in the sign-group, what is delivered in the logogram, how much in the impressions, and before whom in the seal, which identifies the receiving official and not the party (one seal accompanies eight different sign-groups in the archive). Middle Assyrian, which also seals, writes the word *kunukku* 789 times in addition; Crete seals and does not write it.

The cycle does the work of the date. An archive without dates cannot keep multi-annual accounts or justify deliveries to third parties: it records the current cycle. That agrees with what archaeology reports of Minoan archives, which were remade and recycled and not kept from one year to the next, and with Haghia Triada being the photograph of a single year, that of the destruction, written by twenty-one scribes working at the same time. The same asymmetry appears in a different genre and confirms it. The seven perfume recipes of the Assur archive (KAR 140+, KAR 220, KAR 222, PKT 2, 6 and 7, KAL 17,70) write out the whole procedure: light the fire, cover the bowl, let it stand overnight, sieve at dawn, discard the residue, and repeat the pouring while counting the times (the fourth, the seventh, the ninth, the tenth, the eleventh). The Minoan tablets of the Haghia Triada aromatics dossier record the same kinds of ingredient with the same structure of quantities, and write not a single step of the procedure (Acedo 2026e). Assyria writes what is done; Crete, what comes in. The parallel extends to the genre that most resembles it. The Pylos Un tablets record the delivery of ingredients to the perfumer, the same type of document as the Haghia Triada aromatics dossier; PY Un 267 writes the verb of delivery, the name of the deliverer, the name and office of the recipient and the purpose of the delivery, whereas the corresponding Minoan tablets record the same kinds of ingredient with the same structure of quantities and write none of those four things (Acedo 2026e).

6. Discussion

What the comparison allows one to assert. The anatomy of the receipt is universal in the trivial sense: it is the shape any record of deliveries takes, and it appears in administrations with no contact. What distinguishes the traditions is which functions they verbalise, and there Crete does not resemble the Near Eastern model precisely in the two functions that model puts in the foreground. If Minoan bookkeeping had been learned from the Levant or from Mesopotamia, one would expect transfer and date to be written, since they are the most frequent words of the model; their systematic absence points to a native development on a common anatomy.

What it does not allow one to assert. It does not say that Crete was unaware of Near Eastern practice: the Minoans traded with people who dated their tablets and wrote “received”, and they

may have known that practice and not adopted it, which is a decision and not ignorance. Nor does it say anything about language: the form of the document and the language it is written in are independent, and this paper does not touch the second. And it rests on a small Minoan corpus; what is sustained is a claim about what the Haghia Triada archive writes and does not write, and it would be weakened if a Minoan archive with dates were to appear.

What the comparison suggests for the history of trade. The words that did travel to Crete from the East are commodity words (sesame, with its Semitic reduplication preserved; cumin), and those that travelled from Crete to Greece are commodity words too (cyperus, the sycamore fig, perhaps henna). The products and their names travelled, with the ships; the formulary did not. It is an informative asymmetry: Bronze Age trade transmitted goods and lexicon easily and administrative practices with difficulty, which fits the fact that administration is the domain where each palace guarded its own way of doing things.

Two Syro-Levantine archives fall outside for lack of machine-readable data, and this should be stated precisely, since they are the ones closest to Crete. Alalakh (Tell Açana) has 771 objects catalogued in CDLI, 523 of them administrative and 294 from level IV, contemporary with Haghia Triada, but only three with an available transliteration, and those three are lexical texts; its archive is published in the editions of Wiseman and successors, not in tabulated form. Nuzi returns no results in that database under its provenience name. A pending and decisive check is therefore the archive of Ugarit, Crete's direct trading partner in the 14th century. Its corpus is published in non-tabulated editions (the CDLI export gives 2,910 records but only ten with transliteration), and the project that edits it, AEMW, has already published its Amarna letters in open format. With Ugarit measured, the conclusion would move from "the nearest contemporary archive" to "the archive of the trading partner", which is the strong test.

One limitation of this comparison must be declared, since it belongs to the material and not to the method. The archives of Ugarit and Alalakh are treated here from published descriptions and not from the texts, since no digitised corpora of those archives exist in a consultable form comparable to the others (J. Lauinger, personal communication, 9 September 2026). The figures given for Middle Assyrian, by contrast, come from the TCMA corpus and are counts of our own.

Data

TCMA, The Amarna Letters and SAA 7: ORACC, lemmatised JSON format, versions of June and August 2026. Ur III: BDTNS (Base de Datos de Textos Neosumerios) in the redistribution of the COMPASS project. Proto-Elamite and Uruk: CDLI exports of September 2026. Linear A: GORILA (Godart and Olivier 1976-1985) via the LinearA Explorer, with sign verification in SigLA. Scripts: the kuro package.

Declaration of assistance

Corpus processing, the measurements and the first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

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